

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 7 - Formative Assessment 1

Formative Assessment (FA-1) - Answer Key

Class: Class 7

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Networking & Internet, Advanced Graphics

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	b) LAN	LAN connects devices within a single building or small area. WAN covers large geographical areas.
2	c) SMTP	SMTP (Simple Mail Transfer Protocol) is used for sending emails.
3	a) Domain Name System	DNS translates domain names into IP addresses.
4	b) Fuzzy Select Tool	The Fuzzy Select Tool selects contiguous regions of similar color, useful for background removal.
5	d) All layers allow independent editing	In GIMP, each layer can be edited independently without affecting other layers.

Section II: Short Answer Questions (10 marks)

- What is the difference between LAN and WAN? Give one example of each.**
 - Answer:** LAN (Local Area Network) connects devices within a small area like a school or office building. WAN (Wide Area Network) covers large geographical areas like cities or countries.
Example of LAN: Network in a computer lab. Example of WAN: The Internet.
 - Marking:** 1 mark for each correct definition with example (2 marks total).
- Name any four Internet services commonly used today.**
 - Answer:** Email, World Wide Web (WWW), File Transfer Protocol (FTP), Instant Messaging, Video Conferencing, Online Banking.
 - Marking:** 0.5 marks for each correct service (2 marks for four services).
- What is the purpose of a protocol in networking?**
 - Answer:** Protocols are sets of rules that govern how data is transmitted between devices on a network. They ensure that all devices can communicate with each other in a standardized way, preventing errors and ensuring data integrity.
 - Marking:** 1 mark for definition, 1 mark for explaining the purpose.
- Define the term “layer” in image editing. Why are layers useful?**
 - Answer:** A layer is a transparent sheet on which different parts of an image are placed. Layers are useful because they allow editing one part of an image without affecting others, enable non-destructive editing, and make it easy to rearrange or combine elements.
 - Marking:** 1 mark for definition, 1 mark for explaining usefulness.
- What is the difference between a raster image and a vector image?**
 - Answer:** Raster images are made up of pixels and have fixed resolution (e.g., JPEG, PNG).
Vector images are made up of mathematical paths and can be scaled without losing quality (e.g.,

SVG). Raster images are used for photographs, while vector images are used for logos and illustrations.

- **Marking:** 1 mark for each type's description (2 marks total).

Section III: Long Answer Questions (5 marks)

1. **Explain the types of computer networks (LAN, MAN, WAN) with suitable examples and a diagram.**

- **Answer:**

- **LAN (Local Area Network):** Covers a small area like a building or campus. Examples: School computer lab, home network. Features: High speed, low cost, limited range.
- **MAN (Metropolitan Area Network):** Covers a city or large campus. Examples: City-wide network connecting multiple schools. Features: Moderate speed, covers larger area than LAN.
- **WAN (Wide Area Network):** Covers countries or continents. Examples: Internet, bank networks connecting branches. Features: Lower speed than LAN, expensive, unlimited range.

- **Marking:** 2 marks for correct definitions, 2 marks for examples and features, 1 mark for diagram.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Open GIMP and create a new image (800×600 pixels). Add two layers: one for background (fill with a color) and one for drawing a simple shape. Save the file as "LayerDemo.xcf".**

- **Expected:** Image created with correct dimensions, two layers present, background filled with color, shape drawn on second layer, file saved in XCF format.
- **Marking:** 1 mark for creating image with layers, 1 mark for correct save.

2. **Open a web browser and identify the protocol used when visiting "https://www.example.com". List the steps to find the IP address of a website using the command prompt.**

- **Answer:** Protocol used is HTTPS (HyperText Transfer Protocol Secure). Steps: Open Command Prompt → Type nslookup www.example.com → Press Enter → The IP address is displayed.
- **Marking:** 1 mark for protocol identification, 1 mark for correct steps.

3. **Save your GIMP file on the desktop with the correct filename.**

- **Expected:** File saved as "LayerDemo.xcf" on desktop.
- **Marking:** 1 mark for correct filename and location.

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)